# GCSE Mathematics - Foundation Tier

## Paper 3 (Calculator) Practice - Set 9

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

---

### Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use pi = 3.142.

---

**1.**

Write 3/5 as a decimal.

*(Total for Question 1 is 1 marks)*

---

**2.**

Work out 45% of £180.

*(Total for Question 2 is 2 marks)*

---

**3.**

(a) Write 4^3 as an ordinary number. (1)

(b) Write sqrt(81) as an ordinary number. (1)

*(Total for Question 3 is 2 marks)*

---

**4.**

Write the ratio 35 : 49 in its simplest form.

*(Total for Question 4 is 2 marks)*

---

**5.**

The frequency table shows the number of texts sent by 30 students in one day.

| Texts | 0-9 | 10-19 | 20-29 | 30-39 | 40-49 |
|---|---|---|---|---|---|
| Freq. | 4 | 9 | 11 | 4 | 2 |

(a) Write down the modal class. (1)

(b) Work out the percentage of students who sent at least 20 texts. (2)

*(Total for Question 5 is 3 marks)*

---

**6.**

Solve $3(x - 2) = 9$.

*(Total for Question 6 is 2 marks)*

---

**7.**

A right-angled triangle has a hypotenuse of 25 cm and one shorter side of 7 cm.

*[Diagram: right-angled triangle with hypotenuse 25 cm and one shorter side 7 cm; the other side labelled x cm.]*

Work out the length x.

*(Total for Question 7 is 3 marks)*

---

**8.**

The table shows the number of goals scored per match by a netball team over a season.

| | | | | | |
|---|---|---|---|---|---|
| Goals | 20 | 25 | 30 | 35 | 40 |
| Freq. | 3 | 5 | 8 | 6 | 3 |

Work out the mean number of goals scored per match. Give your answer to 2 decimal places.

*(Total for Question 8 is 3 marks)*

---

**9.**

A boat sails from point A on a bearing of 045 deg to point B. Then it sails from B on a bearing of 135 deg to point C.

*[Diagram: point A with north line; AB on bearing 045 deg; B with its own north line; BC on bearing 135 deg.]*

(a) Work out the angle ABC. (2)

(b) Hence describe the triangle ABC. (1)

*(Total for Question 9 is 3 marks)*

---

**10.**

The cost C (in pounds) of running a freezer for d days is given by $C = 0.85d + 12$.

(a) Work out C when d = 30. (2)

(b) How many days does it take to spend exactly £80? (Solve for d.) (1)

*(Total for Question 10 is 3 marks)*

---

**11.**

A cuboid has a square base of side 4 cm and a height of 9 cm.

(a) Work out the volume of the cuboid. (1)

(b) Work out the total surface area of the cuboid. (2)

*(Total for Question 11 is 3 marks)*

---

**12.**

Nadia puts £1500 into a savings account paying 1.5% compound interest per year.

Work out the value of the investment after 4 years. Give your answer to the nearest penny.

*(Total for Question 12 is 4 marks)*

---

**13.**

In a game, a fair coin is tossed and a fair four-sided die is rolled at the same time. The die has faces numbered 1, 2, 3 and 4.

(a) Write down the total number of possible outcomes. (1)

(b) Work out the probability of getting heads on the coin AND a 3 on the die. (3)

*(Total for Question 13 is 4 marks)*

---

**14.**

A right-angled triangle has shorter sides of length 9 cm and 12 cm.

*[Diagram: right-angled triangle with shorter sides 9 cm and 12 cm; hypotenuse labelled.]*

(a) Show that the hypotenuse is 15 cm. (3)

(b) Hence work out the perimeter of the triangle. (2)

*(Total for Question 14 is 5 marks)*

---

**15.**

A piece of metal in the shape of a triangular prism has the following dimensions: cross-section is a right-angled triangle with shorter sides 5 cm and 12 cm; length of prism is 25 cm.

*[Diagram: right-angled triangular prism: cross-section sides 5 cm and 12 cm; length 25 cm.]*

(a) Work out the volume of the prism. (3)

The metal has a density of 4.5 g/cm^3.

(b) Work out the mass of the prism. Give your answer in kilograms. (2)

*(Total for Question 15 is 5 marks)*

---

**16.**

The first four terms of an arithmetic sequence are 4, 9, 14, 19.

(a) Find an expression for the nth term. (2)

(b) Work out the 30th term of the sequence. (2)

*(Total for Question 16 is 4 marks)*

---

**17.**

The table shows the age (years) and resting heart rate (bpm) of 8 adults.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Age | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 55 |
| Rate | 62 | 65 | 68 | 70 | 72 | 75 | 77 | 80 |

(a) On the grid provided, plot these data as a scatter graph. (2)

(b) Describe the correlation. (1)

(c) Draw a line of best fit. (1)

(d) Use your line of best fit to estimate the resting heart rate of a 33 year old. (1)

*(Total for Question 17 is 5 marks)*

---

**18.**

A cylindrical water tank has a radius of 60 cm and a height of 1 m.

(a) Show that the volume of the tank is approximately 1.131 m^3 (use pi = 3.142). (4)

The tank is being filled at a rate of 30 litres per minute (and 1 m^3 = 1000 litres).

(b) Work out the time, in minutes (to the nearest minute), to fill the tank. (1)

*(Total for Question 18 is 5 marks)*

---

**19.**

A scale drawing uses 1 cm to represent 5 km.

(a) A road is 8.5 cm long on the drawing. Work out the real length in kilometres. (2)

Two towns P and Q are 32.5 km apart in real life.

(b) Work out the distance between P and Q on the scale drawing. (2)

(c) Express the scale of the drawing in the form 1 : n. (1)

*(Total for Question 19 is 5 marks)*

---

**20.**

The sum of the first n terms of an arithmetic sequence is given by S_n = n^2 + 2n.

(a) Work out S_3 (the sum of the first three terms). (1)

(b) Work out S_4 (the sum of the first four terms). (1)

(c) Hence work out the 4th term of the sequence. (1)

(d) Show that the sequence has first term 3. (2)

*(Total for Question 20 is 5 marks)*

---

**21.**

A bank offers two savings accounts.

Account X: 2.5% compound interest per year.

Account Y: 2.8% simple interest per year.

Sam invests £4000 in each.

(a) Work out the value of Account X after 5 years. Give your answer to the nearest penny. (3)

(b) Work out the value of Account Y after 5 years. (2)

(c) Which account is worth more after 5 years? By how much? (1)

*(Total for Question 21 is 6 marks)*

---

**22.**

There are 40 chocolates in a box: 18 milk chocolates, 14 dark chocolates and 8 white chocolates.

Tom takes a chocolate at random, notes its type, then puts it back. He then takes a second chocolate at random.

(a) Write down the probability that a single chocolate taken at random is milk. Give your answer in its simplest form. (1)

(b) Work out the probability that BOTH chocolates taken are milk. (2)

(c) Work out the probability that NEITHER chocolate taken is milk. (2)

*(Total for Question 22 is 5 marks)*

---

**TOTAL FOR PAPER: 80 MARKS**

# Mark Scheme

Foundation Tier - Paper 3 (Calculator) - Set 9 | 80 marks

---

**Question 1**

- 0.6 - B1

*Total: 1 marks*

**Question 2**

- 180 x 0.45 or 180 x 45 / 100 - M1
- £81 - A1

*Total: 2 marks*

**Question 3**

- (a) 64 - B1
- (b) 9 - B1

*Total: 2 marks*

**Question 4**

- Both parts divided by 7 (35:49 -> 5:7) - M1
- 5 : 7 - A1

*Total: 2 marks*

**Question 5**

- (a) 20 - 29 - B1
- (b) 11 + 4 + 2 = 17 students; 17/30 x 100 - M1
- (b) 56.7% (accept 56.6 - 56.7 to 1 dp) - A1

*Total: 3 marks*

**Question 6**

- Divide by 3 (x - 2 = 3) OR expand (3x - 6 = 9) - M1
- x = 5 - A1

*Total: 2 marks*

**Question 7**

- x^2 = 25^2 - 7^2 - M1
- 625 - 49 = 576 - M1
- x = 24 (cm) - A1

*Total: 3 marks*

**Question 8**

- Sum(fx) = 60 + 125 + 240 + 210 + 120 = 755 - M1
- Sum(f) = 25 and 755 / 25 - M1
- 30.20 - A1

*Total: 3 marks*

**Question 9**

- (a) Angle from bearing AB (045) plus turn at B to bearing BC (135): supplementary angle to consider - M1
- (a) 90 (deg) - A1
- (b) Right-angled triangle - B1

*Total: 3 marks*

**Question 10**

- (a) 0.85 x 30 + 12 = 25.50 + 12 - M1
- (a) £37.50 - A1
- (b) 80 - 12 = 68; / 0.85 = 80 days - B1

*Total: 3 marks*

**Question 11**

- (a) 4 x 4 x 9 = 144 (cm^3) - B1
- (b) 2(4x4) + 4(4x9) = 32 + 144 - M1
- (b) 176 (cm^2) - A1

*Total: 3 marks*

**Question 12**

- Multiplier 1.015 seen - M1
- 1500 x 1.015^2 (= 1545.3375) seen - M1
- 1500 x 1.015^4 - M1
- £1592.04 (accept £1592.045) - A1

*Total: 4 marks*

**Question 13**

- (a) 8 (= 2 x 4) - B1
- (b) P(heads) = 1/2 and P(3) = 1/4 identified - M1
- (b) 1/2 x 1/4 - M1
- (b) 1/8 - A1

*Total: 4 marks*

**Question 14**

- (a) c^2 = 9^2 + 12^2 - M1
- (a) 81 + 144 = 225 - M1
- (a) c = sqrt(225) = 15 (cm) shown - C1
- (b) 9 + 12 + 15 - M1
- (b) 36 (cm) - A1

*Total: 5 marks*

**Question 15**

- (a) 0.5 x 5 x 12 = 30 (area of cross-section) - M1
- (a) 30 x 25 - M1
- (a) 750 (cm^3) - A1
- (b) 4.5 x 750 = 3375 (g) - M1

- (b) 3.375 (kg) - A1

*Total: 5 marks*

**Question 16**

- (a) Common difference 5 noted; zeroth term -1 - M1
- (a) 5n - 1 - A1
- (b) 5 x 30 - 1 - M1
- (b) 149 - A1

*Total: 4 marks*

**Question 17**

- (a) 8 points plotted correctly - B2 (B1 for 6+ correct)
- (b) Positive correlation - B1
- (c) Sensible line of best fit - B1
- (d) Heart rate at 33 yr (accept 67 - 71 bpm) - B1

*Total: 5 marks*

**Question 18**

- (a) Convert: 60 cm = 0.6 m, height 1 m - M1
- (a) V = pi x 0.6^2 x 1 (= 3.142 x 0.36) - M1
- (a) 1.13112 - M1
- (a) 1.131 (m^3) shown - C1
- (b) 1131 / 30 = 37.7 minutes; rounded 38 minutes - B1

*Total: 5 marks*

**Question 19**

- (a) 8.5 x 5 - M1
- (a) 42.5 km - A1
- (b) 32.5 / 5 - M1
- (b) 6.5 cm - A1
- (c) 1 cm = 5 km = 500 000 cm so 1 : 500 000 - B1

*Total: 5 marks*

**Question 20**

- (a) 3^2 + 2(3) = 9 + 6 = 15 - B1
- (b) 4^2 + 2(4) = 16 + 8 = 24 - B1
- (c) 4th term = S_4 - S_3 = 24 - 15 = 9 - B1
- (d) S_1 = 1 + 2 = 3 - first term equals S_1 - M1
- (d) Therefore first term is 3 - A1

*Total: 5 marks*

**Question 21**

- (a) Multiplier 1.025 seen - M1
- (a) 4000 x 1.025^4 (= 4415.25) seen - M1
- (a) £4525.63 - A1
- (b) 0.028 x 4000 x 5 = 560 - M1

- (b) 4000 + 560 = £4560 - A1
- (c) Y is worth more by 4560 - 4525.63 = £34.37 - B1

*Total: 6 marks*

**Question 22**

- (a) 18 / 40 = 9/20 - B1
- (b) (9/20) x (9/20) or (18/40)^2 - M1
- (b) 81/400 (accept 0.2025) - A1
- (c) P(not milk) = 22/40 = 11/20; (11/20)^2 - M1
- (c) 121/400 (accept 0.3025) - A1

*Total: 5 marks*

# Worked Solutions

Foundation Tier - Paper 3 (Calculator) - Set 9 | Detailed explanations

---

**Question 1**

3/5 = 3 / 5 = 0.6.

---

**Question 2**

45% of 180 = 0.45 x 180 = 81.

Answer: £81.

---

**Question 3**

(a) 4^3 = 4 x 4 x 4 = 64.

(b) sqrt(81) = 9 (because 9 x 9 = 81).

---

**Question 4**

HCF of 35 and 49 is 7.

35/7 = 5, 49/7 = 7. Answer: 5 : 7.

---

**Question 5**

(a) Highest frequency is 11, in the 20-29 group. Modal class: 20-29.

(b) "At least 20" = 11 + 4 + 2 = 17 students.

Percentage = 17 / 30 x 100 = 56.66...%, i.e. 56.7% to 1 dp.

---

**Question 6**

3(x - 2) = 9. Divide by 3: x - 2 = 3. Add 2: x = 5.

Check: 3(5 - 2) = 3 x 3 = 9.

---

**Question 7**

Pythagoras for the shorter side: x^2 = 25^2 - 7^2 = 625 - 49 = 576.

x = sqrt(576) = 24 cm.

---

**Question 8**

Sum(fx) = 20x3 + 25x5 + 30x8 + 35x6 + 40x3 = 60 + 125 + 240 + 210 + 120 = 755.

Sum(f) = 3 + 5 + 8 + 6 + 3 = 25.

Mean = 755 / 25 = 30.20.

---

**Question 9**

(a) At B, two bearings meet: the back-bearing of AB at B is 045 + 180 = 225 deg, and the new bearing BC is 135 deg.

Angle ABC (the interior angle of triangle ABC at B) = 225 - 135 = 90 deg.

(b) Since angle ABC = 90 deg, triangle ABC is a right-angled triangle (right angle at B).

---

**Question 10**

(a) C = 0.85 x 30 + 12 = 25.50 + 12 = 37.50.

Answer: £37.50.

(b) Solve 0.85d + 12 = 80. So 0.85d = 68 and d = 80 days.

---

**Question 11**

(a) V = 4 x 4 x 9 = 144 cm^3.

(b) Top and bottom: 2(4x4) = 32. Four side rectangles: 4(4x9) = 144. Total = 176 cm^2.

---

**Question 12**

Multiplier 1.015.

After year 1: 1500 x 1.015 = £1522.50.

Year 2: 1522.50 x 1.015 = £1545.3375.

Year 3: 1545.3375 x 1.015 = £1568.5176...

Year 4: 1568.5176... x 1.015 = £1592.0454...

Or 1500 x 1.015^4 = 1500 x 1.061363... = £1592.0454...

To the nearest penny: £1592.04.

---

**Question 13**

(a) 2 outcomes for coin x 4 for die = 8.

(b) Coin and die are independent.

P(heads) = 1/2. P(die shows 3) = 1/4.

P(both) = 1/2 x 1/4 = 1/8.

---

**Question 14**

(a) c^2 = 9^2 + 12^2 = 81 + 144 = 225.

c = sqrt(225) = 15. (Shown.)

(b) Perimeter = 9 + 12 + 15 = 36 cm.

---

**Question 15**

(a) Area of cross-section = 0.5 x 5 x 12 = 30 cm^2.

Volume = 30 x 25 = 750 cm^3.

(b) Mass = 4.5 x 750 = 3375 g = 3.375 kg.

---

**Question 16**

(a) Common difference 5. Zeroth term = 4 - 5 = -1. nth term = 5n - 1.

(b) 30th term = 5(30) - 1 = 149.

**Question 17**

(a) Plot 8 points.

(b) Positive correlation.

(c) Best fit drawn.

(d) At 33 years, line gives roughly 67 - 71 bpm.

**Question 18**

(a) Radius 60 cm = 0.6 m; height 1 m.

V = pi x r^2 x h = 3.142 x 0.6^2 x 1 = 3.142 x 0.36 = 1.13112.

To 3 dp: 1.131 m^3. (Shown.)

(b) 1.131 m^3 = 1131 litres. Time = 1131 / 30 = 37.7 minutes, i.e. 38 minutes.

**Question 19**

(a) 8.5 cm x 5 km/cm = 42.5 km.

(b) 32.5 km / 5 km/cm = 6.5 cm.

(c) 5 km = 500 000 cm. So 1 cm represents 500 000 cm, i.e. scale is 1 : 500 000.

**Question 20**

(a) S_3 = 3^2 + 2(3) = 9 + 6 = 15.

(b) S_4 = 4^2 + 2(4) = 16 + 8 = 24.

(c) 4th term = S_4 - S_3 = 24 - 15 = 9.

(d) The first term equals S_1 = 1^2 + 2(1) = 1 + 2 = 3.

So the first term is 3.

**Question 21**

(a) Account X (compound): 4000 x 1.025^5.

1.025^2 = 1.050625; 1.025^4 = 1.103813...; 1.025^5 = 1.131408...

Value = 4000 x 1.131408... = £4525.63.

(b) Account Y (simple): interest per year = 0.028 x 4000 = £112. Over 5 years: 112 x 5 = £560.

Total = 4000 + 560 = £4560.

(c) Y is worth more by 4560 - 4525.63 = £34.37.

**Question 22**

(a) P(milk) = 18 / 40 = 9/20.

(b) With replacement, the two picks are independent.

P(both milk) = (9/20) x (9/20) = 81/400 = 0.2025.

(c) P(not milk) = 22 / 40 = 11/20.
P(both NOT milk) = (11/20) x (11/20) = 121/400 = 0.3025.